

The XELOR agent system

Seven agents, sixteen tools and five missions — and the machinery that keeps all of it inside the authority of the person who asked.

Covers: what XELOR is, the seven agents, the architecture, every capability, all five mission graphs, how a run actually executes, the trust model, and what is honestly not built.

Companion set: one guide per agent.

Truth standard: the repository as it stands on 3 August 2026.

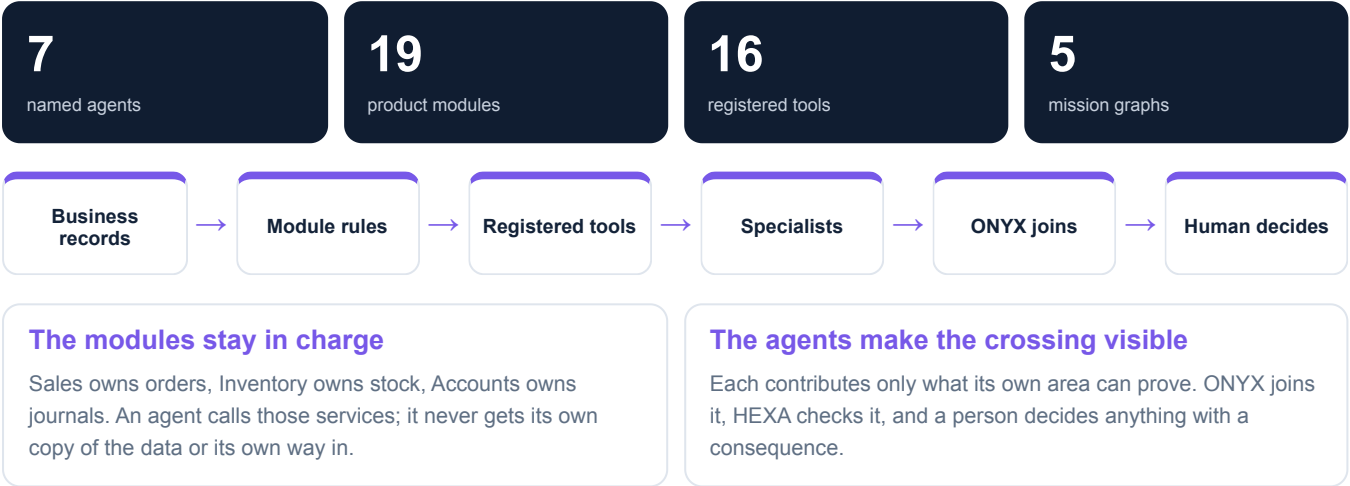
MASTER

What XELOR is

A manufacturing system of record, with a controlled layer that reads across it

The plain version

A factory's information sits in separate places — orders, stock, plans, the shop floor, quality, machines, money, people. Each has its own rules and each is right about its own subject. What nobody has is one honest answer to a question that crosses all of them. XELOR's agents exist for that question, and for nothing else: they do not replace the modules, they read them under the asker's own permissions and put the pieces side by side.



Seven agents, one system

Each has a full companion guide

ONYX · Supervisor

Turns a business goal into a bounded plan, assigns the right specialists, joins their evidence and presents one understandable result.

HEXA · Governance

Holds identity, tenant isolation, permissions, approvals and the tamper-evident record, so every mission stays inside the authority of the person who started it.

MICA · Commercial

Explains what a customer ordered, what was promised, what has shipped, and what service or warranty obligation is still open.

SPAR · Supply

Explains what stock exists, what has been ordered from suppliers, and where a shortage threatens a plan or a promise.

AXLE · Planning

Connects what the product is made of with what demand requires, then says what to make, what to buy, by when — and which dates cannot be met.

KILN · Operations & Quality

Explains what the plant is making, whether it passes inspection, and whether the machines will still be there tomorrow.

RASP · Finance & People

Explains the accounting truth, the cash position, what is already committed, and the payroll consequence behind an operating decision.

The counterintuitive part, and the best proof the design is real

ONYX is the supervisor, and it is the LEAST powerful agent in the system. It holds one read tool and is not on the allow-list for the one capability that can cause an effect. It can convene a mission and write the summary; it cannot touch the business. A design where the coordinator accumulates power is the failure everyone expects from agent systems — this one gives it the smallest surface of all.

How authority is distributed

One supervisor

ONYX is the only agent that may delegate. The six specialists cannot recruit each other or widen their own role.

One control layer

HEXA verifies before and after the human gate, and applies the same tenant, permission and approval rules to everybody.

Nineteen modules under seven owners

The master gives the map; the agent guides explain each module in depth

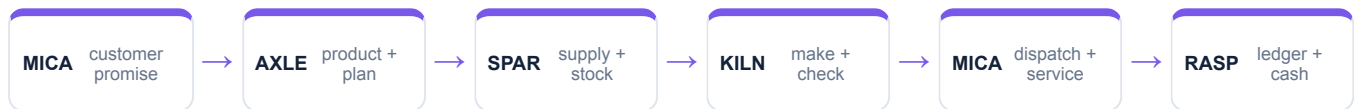
Agent	Modules	What the group contributes
ONYX Supervisor	ONYX Copilot · Agent OS · AI Operations	Copilot handles one bounded question; Agent OS handles a cross-functional mission; AI Operations governs any model-assisted feature. They share identity, permissions, logging and refusal rules, but they do not share a hidden general-purpose action channel.
HEXA Governance	Organisation · Administration · Integration	Organisation supplies the legal and numbering context, Administration decides who may do what and proves the trail, and Integration carries external failures as explicit state. Together they are the control plane used by every business module and every agent run.
MICA Commercial	Sales · Service	Sales records what was promised and fulfilled; Service records what happened after delivery. MICA links the same customer, order, installed-base and quality evidence without allowing the portal to cross into another customer's account.
SPAR Supply	Purchase · Inventory	Purchase creates an approved supply commitment; Inventory records the physical consequence. A receipt is useful only when both succeed in one transaction, and downstream modules move stock through Inventory rather than maintaining their own balances.
AXLE Planning	Engineering · Planning	Engineering defines what a product is; Planning applies demand, stock, supply, calendars and policy to decide what must happen by date. Production receives proposals and a pinned structure, so later master-data changes cannot rewrite work already released.
KILN Operations & Quality	Production · QMS & Audit · Maintenance	Production records the work, Quality decides whether the result is acceptable, and Maintenance explains whether the equipment can sustain the plan. Inventory movements are shared evidence, while quality and downtime conclusions remain deterministic and human-accountable.
RASP Finance & People	Accounts · Employee Spend · People · Working Capital	Accounts is the posted truth; Employee Spend and People create controlled obligations and postings; Working Capital turns the available evidence into reviewable priorities and scenarios. RASP must label the difference between a ledger fact, an assumption and an illustrative demo value.

One ownership rule prevents a great deal of confusion

The module that owns a record is the only place allowed to change it. A mission can combine Sales, stock, plan, quality and finance evidence, but it does not create a second cross-functional copy. The agent layer coordinates decisions; domain services remain the system of record.

How the modules complete one factory story

Control and evidence travel with the transaction



1 · Commit

Sales stores the dated customer promise, GST result and credit decision. Planning receives demand; it does not reinterpret the order.

2 · Plan and supply

Engineering defines the BOM. Planning nets demand against stock and supply. Purchase approves a commitment; Inventory records the physical receipt.

3 · Execute and prove

Production pins components and operations. Quality snapshots the limits it judged. Maintenance preserves the uptime evidence behind delivery risk.

4 · Deliver and account

Dispatch posts stock-out and the invoice together. Accounts holds the receivable and later receipt; Service keeps the after-delivery obligation.

HEXA surrounds every step

Verified identity, tenant fence, permission, approval and audit apply before the business service is allowed to touch a record.

ONYX crosses the loop safely

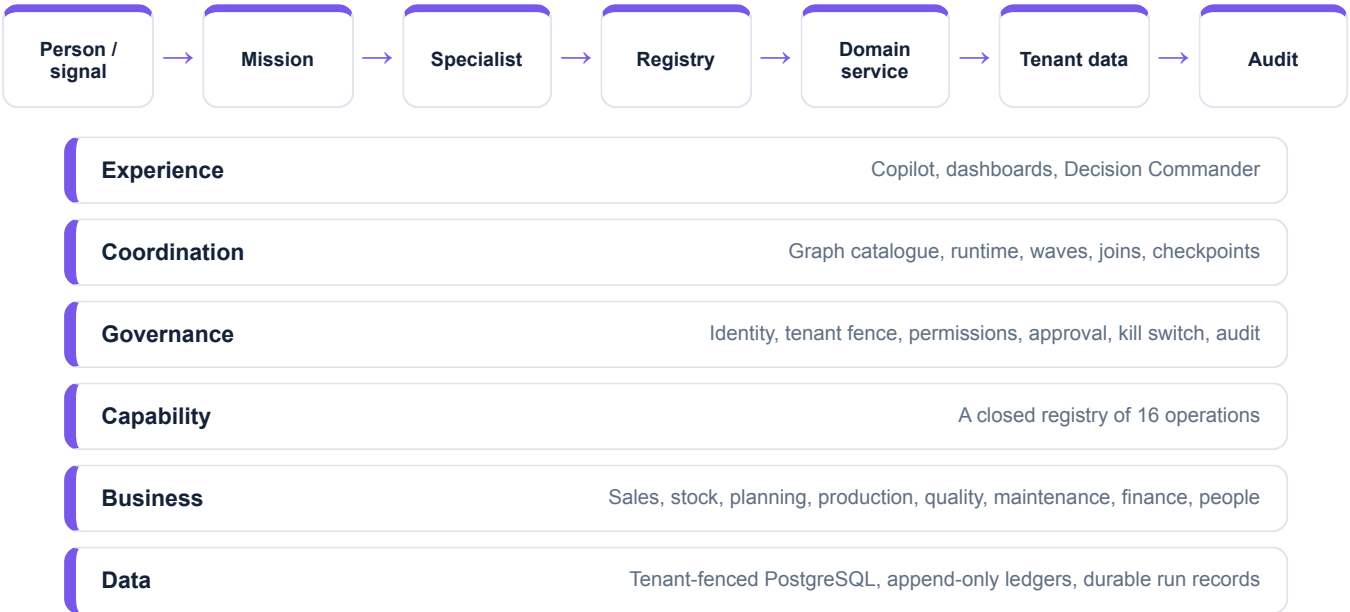
It asks the six specialists for evidence, joins their answers and waits for the person. It does not take ownership away from any module.

The Northstar proof

The canonical demo follows one 120-unit PX-400 customer order through MRP, purchase, three production orders, a rejected inspection, 12 quarantined units, 28 dispatched units, an invoice, a receipt, a service ticket, employee spend and a Decision Commander approval. The same identifiers reconcile across the modules.

How a question reaches trusted data

Every layer narrows what is possible and keeps the evidence



There is no database agent

An agent cannot write a query. It names a registered capability; the runtime checks who is asking and whether they may; the capability calls a service whose rules are already in force. The reason the Copilot can refuse “ship the held units anyway” is not good judgement — it is that no endpoint exists that would take such an instruction.

All 16 capabilities

The complete registry — nothing else is callable

Capability	Mode	Who may call	What it returns	Permission
general.companies.read	Read	HEXA, ONYX	Company masters	general.company.read
sales.orders.read	Read	MICA	Sales orders	sales.order.read
inventory.on-hand.read	Read	SPAR	Stock balances	inventory.stock.read
planning.planned-orders.read	Read	AXLE	Planned orders	planning.mrp.read
production.orders.read	Read	KILN	Production orders	production.order.read
quality.inspections.read	Read	KILN	Inspections	quality.inspection.read
accounts.vouchers.read	Read	RASP	Journal vouchers	accounts.ledger.read
finance.cash-position.read	Read	RASP	Trial balance	accounts.ledger.read
quality.evidence.collect	Analyse	KILN	Evidence view	quality.inspection.read
finance.collections.prioritise	Analyse	RASP	Review queue	accounts.ledger.read
finance.forecast.simulate	Simulate	RASP	13-week scenario	accounts.ledger.read
quality.audit-plan.draft	Draft	KILN	Audit checklist	quality.inspection.read
quality.capa-plan.draft	Draft	KILN	Investigation draft	quality.inspection.read
quality.audit-pack.draft	Draft	KILN	Evidence manifest	quality.inspection.read
finance.funding-pack.draft	Draft	RASP	Draft manifest	accounts.ledger.read
agent.action.dispatch	Execute	HEXA, MICA, SPAR, AXLE, KILN, RASP	Governed work item	agentos.run.operate

Fifteen of sixteen cannot change anything

Exactly one capability has a side effect, and a test asserts that count so it cannot drift. That one requires an approved human gate above it in the graph, and the engine checks the ancestry itself rather than trusting the capability to behave. Note also that ONYX is absent from the execute row.

Every registered graph

69 nodes in total, all written down before any run starts

Cross-functional readiness review

foundation.cross-functional-readiness · 9 nodes · step budget 14 · timeout 300s · 0 dispatch nodes

ONYX frames it · MICA and SPAR read in parallel · both assess · join · HEXA verifies · human approves · ONYX synthesises.

Seven-agent operating review

operations.full-command-review · 17 nodes · step budget 24 · timeout 300s · 0 dispatch nodes

All six specialists read their own domain at once, each assesses, ONYX joins, HEXA verifies four checks, a human approves, ONYX issues the brief.

Seven-agent controlled action mission

operations.controlled-action-mission · 26 nodes · step budget 32 · timeout 600s · 6 dispatch nodes

The read-propose-verify-approve-dispatch-verify contract. Six reads, six recommendations, one plan, a preflight, ONE human gate, then six governed work items and an outcome check.

Working Capital Review

finance.working-capital-review · 9 nodes · step budget 18 · timeout 300s · 0 dispatch nodes

RASP reads the trial balance, MICA the commitments, SPAR the stock holding; RASP simulates 13 weeks; HEXA verifies; a human approves the brief.

QMS & Audit Readiness

quality.qms-audit-readiness · 8 nodes · step budget 18 · timeout 300s · 0 dispatch nodes

KILN reads inspections, collects evidence and drafts a pack; HEXA checks traceability; the quality owner approves; ONYX publishes gaps and owners.

Four of the five cannot act at all

Only the controlled action mission contains dispatch nodes, and it has six — one per specialist, every one of them structurally downstream of a single human gate. The other four graphs are read, analyse, draft and explain. A test asserts that the Working Capital and QMS missions contain no dispatch node, so that property cannot be edited away by accident.

The execution rules, in plain terms

The same engine drives all five missions

1

Everything ready runs together

The engine takes every node whose dependencies are finished and runs that whole wave at once. Six specialists reading six modules is one wave, not six trips.

2

A checkpoint after every wave

Progress is written down as it happens, so a run can be explained afterwards and resumed rather than restarted.

3

Retries are bounded

A node may be given a second attempt, never more than five, and the limit is validated when the graph is registered rather than at runtime.

4

Failure stops the mission

A failed node, a timeout, or a state where nothing can progress ends the run and says so. It does not invent a result to keep things moving.

5

Interrupted work is recovered honestly

If the process dies mid-node, that node is returned to pending on resume and re-run — with its attempt count preserved, so recovery cannot be used to escape the retry budget.

6

Approval pauses everything

An approval node parks the run. The decision belongs to that exact node in that exact mission and cannot be reused elsewhere.

7

Effects need an approved ancestor

Before a side-effecting node runs, the engine walks the graph upward looking for an approval that actually returned yes. No ancestor, no execution — checked by the engine, not by the tool.

Why the person stays in control

Five boundaries, on every mission, with no exceptions

1 · Identity

A verified token decides the tenant and the actor

2 · Authority

The agent's allow-list AND the person's own permission

3 · Action

A side effect needs an approved ancestor node

4 · Business rules

The normal domain service still decides what is legal

5 · Evidence

Runs, nodes, events, checkpoints and audit all persist

Failure is visible

A failed node, a timeout or a deadlock is recorded and the run stops. Nothing is smoothed over to keep a demo comfortable.

Recovery is real

Checkpoints let an interrupted run continue without pretending the interrupted step had finished.

AI is fenced separately

A closed registry of 8 features, a kill switch, budgets and an evaluation gate that must beat a rules-only baseline before anything ships.

Code decides, people approve

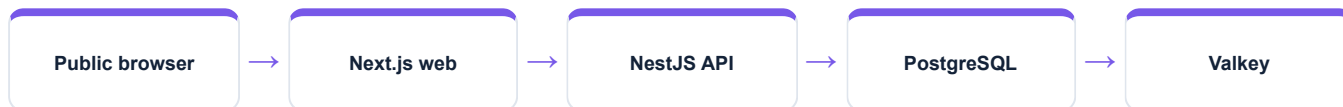
Tax, stock, quality verdicts, payroll and accounting are arithmetic. A model may phrase an explanation; it never sets the number.

Where the guarantees are actually kept

Continuous integration re-proves the tenant fence on every build, a live two-tenant probe tries to read across the boundary and fails, and all 138 permissions are reconciled against the routes that demand them. These are gates, not intentions.

How the shareable build is packaged

Railway-ready five-service topology; no live cloud URL is claimed



Web service

Dynamic Railway port · routes API calls to the public API URL

API service

Waits for PostgreSQL · migrates · seeds/checkpoints once · starts HTTP

Worker service

Consumes the transactional outbox independently from the API

PostgreSQL

Custom demo image with extensions, application role and all durable records

Valkey

Redis-compatible cache/coordination service, not the source of business truth

Public-demo identity

The hosted demo opens without the former sign-in screen. An explicit public-demo mode maps a small, fixed persona list to seeded tenant permissions; normal bearer-token checks remain the default outside that mode.

Repeatable boot

Deployment scripts wait for dependencies, apply all migrations, seed only when the checkpoint is absent, and expose liveness/readiness endpoints. Restarts do not blindly rebuild the database.

Current status

The repository contains Railway Dockerfiles, service descriptors, a Compose-equivalent topology and a deployment runbook. A production domain or confirmed public Railway URL is not present in the working directory, so the honest state is “deployment-ready demo packaging”, not “currently hosted”.

What this build does and does not do

Say the right-hand column out loud before anyone finds it

Live in the repository

- Seven agents with fixed allow-lists
- 16 registered capabilities, one of them effectful
- Five versioned, content-hashed mission graphs
- Parallel waves, checkpoints, bounded retries, recovery
- Tenant fencing and permissions, proven by CI
- Approval gates with an append-only trail
- Idempotent, approval-bound work-item dispatch
- Hash-chained audit with a real verifier

Not claimed

- No external model API is active — reasoning is deterministic
- No external connector is live; integration is simulate-driven
- No general SQL tool, and no way to create one at runtime
- No agent can widen the asking person's permissions
- No automatic customer message, payment or posting
- A dispatched work item is never described as completed work
- No quotation, lead or engineering change control exists
- The purchase cycle stops at the goods receipt

The one-line disclosure

Orchestration, ERP reads, approval gates and governed dispatch are live. Language reasoning is deterministic, and no external model API or connector is active.

Where to read further

The master stays short; each agent has its own handbook

ONYX · Supervisor ONYX Copilot, Agent OS, AI Operations	HEXA · Governance Organisation & master data, Administration, Integration
MICA · Commercial Sales, Service portal (CSP)	SPAR · Supply Purchase, Inventory
AXLE · Planning Engineering, Planning	KILN · Operations & Quality Production, Quality & QMS, Maintenance
RASP · Finance & People Accounts, Employee spend, People, Working capital	

Each guide follows the same twelve pages: purpose, two module deep dives, the end-to-end business pipeline, enforced rules, runtime contract, exact tools, mission participation, a worked example, honest limits and a quick reference.

Shared glossary

Agent	A named specialist with a fixed, allow-listed tool surface.
Capability	One registered operation with a mode, an owner and a required permission.
Mission graph	A versioned recipe for steps, parallel work, joins and approval gates.
Checkpoint	A durable progress snapshot used for explanation and recovery.
Governed work item	An approved, append-only assignment — not the business change itself.
Verified value	An observed result with its method recorded, kept distinct from an estimate.

Sources: the capability registry, graph catalogue and run engine; the module services and their migrations; the platform packages for tax, quality, planning, accounting and audit; the automated tests; and the project's own investor-demo capability-gap register. Snapshot: 3 August 2026.